Dhaba Prep Assistant

Product Development Assignment Final Submission

> **Executive One-Line Scope**

1. Project Submission Metadata

Submission Item	Details / Reference Link
Product Name	Dhaba Prep Assistant MVP
Target User	Dhaba & Cafe Owners / Line Head Cooks (Single
Live Link to MVP	https://dhaba-prep-mvp.web.app (or local :808
GitHub Repository	https://github.com/Prakalp-ux/dhaba-prep-mvp.
Git Commit Hygiene	11 Incremental Feature Commits (Strict Sequen
Core Tech Stack	Mobile-first Plain HTML5, CSS3, ES6 JS, Fireb
2-Min Demo Video	Recorded via Live Link / Localhost (Script en
Hallway Usability Test	Completed with Quick-Service Cafe Operator (N

2. Core User Journey

The target user is a busy dhaba owner or head cook with dusty or wet hands who needs high contrast, large buttons (48px+), and zero typing friction.

1. End-of-Day Food Logging:

At closing (under 3 mins), enters Made, Left, and Wasted quantities for core prep items: Dal, Paneer, Mixed veg, and Rice.

2. Tomorrow’s Prep Planning:

Next evening taps Tomorrow’s prep to see data-driven cooking recommendations based on the 7-day average plus a 10% safety buffer.

3. Shift Handover & Morning Check:

Closing staff logs physical stock (Milk, Paneer, Eggs, Bread) with a note. Morning shift sees differences (e.g. -2 L) instantly.

4. Party / Advance Demand:

Records booked catering orders (head count, advance paid) that automatically increase prep recommendations.

3. Feature Checklist (MoSCoW Prioritization)

Every single candidate feature was evaluated against user interview evidence. Only Must-haves were built; all others were cut to preserve MVP discipline.

Feature Name	MoSCoW	Status	Rationale
End-of-day stock & waste log	Must	DONE	Core spillage & inventory mismatch problem (P
Tomorrow’s prep suggestion	Must	DONE	Prevents blind over/under-cooking and 15-min
Closing count & handover note	Must	DONE	Solves night-shift staff shrinkage and missin
Party / advance order entry	Must	DONE	Predictable demand feeds extra cooking volume
Waste value in rupees	Should	CUT	Needs pricing models and dish costing; adds m
Recipe & staff training cards	Should	CUT	Cooks know recipes; primary bottleneck is bat
WhatsApp summary export	Should	CUT	Excluded in Section 1.5 to maintain standalon
Hindi / regional language UI	Should	CUT	Universal icons and simple English words suff
POS / token ordering	Won’t	CUT	Completely different product requiring paymen
Staff theft CCTV AI	Won’t	CUT	Sensitive and expensive; physical count provi

4. The Suggestion Rule (Mathematical Specification)

The engine implements the exact mathematical rule defined in Section 1.8 of the product brief with zero invented heuristics:

Daily Consumption Formula:

For each item on day d: Sold = Made - Left - Wasted. (Clamped to 0 if negative).

Rolling Average:

Averages the Sold values across the most recent 7 available days (prior to tomorrow).

Safety Buffer & Rounding:

Suggested = Average * (1 + BUFFER) with BUFFER = 0.10, rounded to nearest 0.5 kg/L.

Transparent Confidence Badges:

Displays ’Based on last 7 days’ (N>=7), ’Based on N day(s) of logs’ (1<=N<=6), or ’No logs yet - enter a quantity yourself’ (N=0).

Party Extra Addition:

Sums quantities from partyOrders dated tomorrow: Total to cook = Adjusted + Party Extras.

5. Code Architecture & Plain-Language Notes

Prepared for assignment review and oral defense. Explains every file in plain language with zero technical jargon.

File Name	Role in App	Key Functions	What Breaks if Deleted
index.html	Single-Page UI Skeleton	Holds screens A, B, C, D, D2, E	No visual UI renders in browser
css/styles.css	Mobile-First Styling	48px+ touch targets, dhaba palette	Unstyled, unusable tiny links
js/firebase-config.js	Cloud DB Credentials	firebaseConfig object, config check	Database connection fails to init
js/db.js	Data Read/Write Layer	saveDayLog, savePrepPlan, getLogs	Zero persistence; app cannot save
js/suggestion.js	Mathematical Intelligence	calculatePrepSuggestion, roundToHalf	No prep suggestions; cooks guess blind
js/app.js	Screen & UI Controller	showScreen, validateInputs, steppers	Buttons dead; dynamic forms frozen
firestore.rules	Cloud Security Access	Restricts read/write to 6 collections	Database vulnerable or test-mode expires

Worked Numerical Example for Oral Defense

Suppose Paneer sold over the last 3 days was: Monday 4.0 kg, Tuesday 4.5 kg, and Wednesday 3.5 kg.

Step 1: Calculate Average:

Sum = 4.0 + 4.5 + 3.5 = 12.0 kg. Average = 12.0 / 3 = 4.0 kg.

Step 2: Apply 10% Safety Buffer:

4.0 kg * 1.10 = 4.4 kg.

Step 3: Round to Nearest 0.5 kg:

4.4 kg rounds to 4.5 kg. Badge shows: 'Based on 3 day(s) of logs - not enough yet'.

Step 4: Party Extra Demand:

If an upcoming party order booked 2.0 kg Paneer tomorrow, Total to cook = 4.5 + 2.0 = 6.5 kg.

6. Comprehensive Build Log (Git History & Fixes)

Chronological log tracking each iterative prompt, what functioned on first run, what broke, and how it was resolved.

Commit Hash	Session & Task	What Broke / Challenge	Resolution Applied
b4ce8d1	Step 1: Git & Brief	Out-of-scope files from previous draft	Pruned wrong files; wrote authentic brief
e083f09	Step 2-3: Plan & Scope	Potential feature creep (rupee waste)	Enforced strict 5-task build order
93eaf59	Step 4: Task 0 Skeleton	Firebase placeholder credentials error	Created fallback adapter in js/db.js
54bce36	Step 6: Task 1 EOD Log	Negative numbers typed on touch keypad	Added inline red warnings blocking save
322c787	Step 7: Task 2 Party Orders	Submitting empty food item lists	Mandated at least 1 item with qty > 0
ba1e60b	Step 8: Seed Test Data	Testing suggestions without 7 real days	Created standalone unlinked seed.html
3015191	Step 9: Task 3 Prep Engine	Stepper allowed decreasing below 0 kg	Added min 0 bounds check on adjusted values
f40b406	Step 10: Task 4 Closing	Floating-point discrepancies (e.g. -0.1999)	Passed diffs through roundToHalf()
171f3b4	Step 11: Phone Sweep	Party select row cramped on 360px width	Adjusted flex ratio (flex 2 on dropdown)
ee49942	Step 12: Code Notes	Database terminology too academic	Rephrased into kitchen shift analogies
280188b	Step 16: Hallway Test Fix	Tester confused if stepper was base or total	Added 'Base Prep Target' & 'Total to cook'

7. Hallway Usability Test & Iteration

Tested on an Android smartphone with Rohan, a cafe operator who had never seen the app before.

Instruction Given:

'You run a small cafe. Log today's food and see tomorrow's prep.' (Observer stayed silent).

Observed Hesitation:

Tester smoothly logged daily food but hesitated at Tomorrow's prep: 'Does 4.5 kg mean total food for tomorrow or what I need to cook extra?'

Smallest Change Implemented:

Added sub-header 'Base Prep Target' above steppers and re-labeled bottom line to 'Total to cook tomorrow:'.

Retest Outcome:

Tester immediately understood the distinction and completed the end-to-end task in under 2 minutes.

8. 2-Minute Screen-Recorded Demo Script

Follow this second-by-second timeline when recording your screen demo:

Time Window	On-Screen Action	Word-for-Word Voiceover Script
0:00 - 0:10	Today screen on phone	'This is Dhaba Prep Assistant. It solves one
0:10 - 0:40	Tap End-of-day log	'At closing, the cook enters Made, Left, and
0:40 - 1:05	Tap Party orders	'If a party of 20 is booked for tomorrow, the
1:05 - 1:35	Tap Tomorrow's prep	'The app calculates prep using a 7-day averag
1:35 - 1:55	Closing count & Handover	'Closing staff counts stock and leaves a note
1:55 - 2:00	Live link in browser bar	'This is completely live, mobile-first, and w